

in the market would cost you over Rs 1,00,000.” The team managed to keep the price low by using low-cost hardware components that were readily available. Another cost-cutting factor is that the project runs on free and open source software, mostly written in the Python programming language.

One of the unique features of Phoenix is its job division. The microcontroller collects real-time data from the experiment, while the software running on the PC takes care of complex processing and graphical representation of data on the computer screen.

Phoenix’ hardware consists of an Atmel ATmega32 microcontroller running a ‘C’ program, a 12-bit resolution analogue-to-digital converter (ADC), a 12-bit digital-to-analogue converter, variable gain amplifiers, several oscillators and a programmable constant-current source. It is interfaced to a computer that runs Python.

“For the software, we have chosen Python, as it has a huge collection of libraries and programs for scientific computation. Several GUI programs have already been written to perform various experiments. A user can conduct several experiments using the GUI programs provided, but designing new experiments would require accessing the hardware accordingly. Hardware features can be accessed by entering codes that are well documented from the Python library,” Dr Kumar explains.

Phoenix’s clear-cut architecture is one of its USPs, enabling teachers to develop new experiments with the help of the library. Dr Kumar elaborates, “School teachers are usually not inclined to learn about electronics or micro-level C-programming, but this is usually required when communicating with the hardware. With Phoenix, all they need to do is write one line of code that is readily available in the Python Library, and they will get the required data. For example, if you want to capture the frequency from a guitar, you can type a code on the computer and send a request to the hardware unit. The computer’s monitor will display the frequency while you remain uninvolved with the programming details.”

The communication takes place between the microcontroller in the hardware unit and the computer running Python through a USB connection.

Dr Kumar adds, “The earlier units that we designed had parallel ports but this kind of interfacing lost popularity. Hence we migrated to the USB interface. It also helped that Python already had a library to communicate to the serial port.”

Overcoming challenges

Ensuring a low price point for Phoenix was one of the biggest challenges before Dr Kumar and his team while designing the platform. Getting high-resolution components at low prices was a challenge in itself and required a lot of research. For example, the IUAC team went to different markets to compare prices, easy availability and resolutions before selecting ADCs. They decided to use a product from Microchip which was available for Rs 150.

“Apart from affordability, we looked for components that



expEYES hardware

could be sourced easily and repaired by the users in case of damages that are common in school and college laboratories,” reasons Dr Kumar.

The IUAC team also worked on the flexibility of the framework by writing the software in a generic manner. It is easy to get the hardware and write a simple software that makes the platform rigid. However, this would have meant only a limited number of experiments that could be done on the platform.

Dr Kumar illustrates, “We can easily design platforms that work for five experiments, but for the sixth one, the user should know all the details of hardware design and coding. This makes the architecture rigid. In Phoenix, a lot of time and effort went into making the architecture flexible. Its generic nature allows software that supports time measurement, to be used to measure acceleration due to gravity and the velocity of sound.”

What lies ahead

IUAC’s goal is to help engineering students develop their own projects rather than purchase them from professionals in the field. It has approached NCERT and CBSE to make it a part of every school’s lab activities.

“As of now, the only explanation given for AC and DC is that AC means alternating current and DC means direct current. Most of the teachers cannot answer a simple query like, “What is the nature of voltage on a three-point mains socket?” We hope that Phoenix will help them improve by changing their practices, and making students and teachers more aware of what is in their books,” Dr Kumar concludes. 

Reference:

[1] <http://expeyes.in>

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